Data Sheet

EXPERIMENT NO. 06

Title: - Simulation of Inverting and Non-Inverting Amplifier using Op-Amp (using Tinkercad/ Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	Signal Generator		1
2.	Oscilloscope		1
3.	Op-amp		1
4.	Resistance		(2/3)
5.	Ground		1
6.	Connecting Wires		require

Important Formulae: -

Inverting Amplifier: -

$Vo / Vi = - Rf / R$

$Vo = - (Rf / R) \times Vi$

Non-Inverting Amplifier: -

$Vo / Vi = 1 + (Rf / R)$

$Vo = [1 + (Rf / R)] \times Vi$

- Vi = Input voltage
- Vo = Output voltage
- R = Input resistance
- Rf = Feedback resistance

Observation Table: -

Table 1: Inverting Amplifier — $Vo = - (Rf / R) \times Vi$

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Title :- Simulation of Inverting and Non-Inverting Amplifier Using Op-Amp (Using Tinkercad / proteus Software Circuit).

Theory :- Operational amplifiers can be configured with external feedback resistors to create precise, stable voltage amplifiers. In an inverting configuration, the input signal is applied to the negative terminal, resulting in an amplified output that is 180 deg out of phase with the input. In a non inverting configuration the signal is applied to the positive terminal, producing and amplified output that remains in phase with the input, defined by the formula $V_0 = [1 + (R_f/R)] \times V_i$

Inverting Amplifier :-

$V_0/V_i = - R_f/R$ $\quad V_0 = - (R_f/R) \times V_i$

Non Inverting Amplifier :-

$V_0/V_i = 1 + (R_f/R)$ $\quad V_0 = [1 + (R_f/R)] \times V_i$

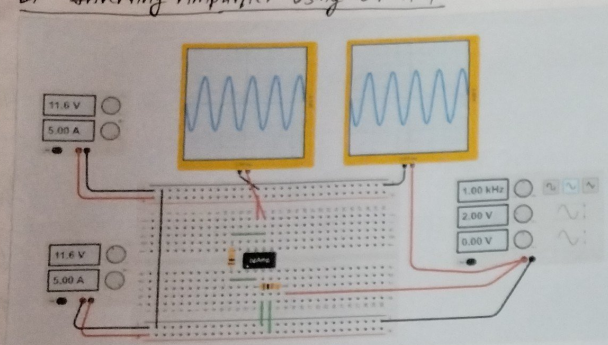
Observation Table :- (Inverting Amplifier)

Sl.no	R_f	R	V_i	V_0 (Calculated)	V_0 (Measured)
1.	10	2.2	0.20	- 0.409	- 0.005
2.	15	3.3	0.25	- 1.136	- 1.192
3.	22	4.7	0.30	- 1.404	- 1.397

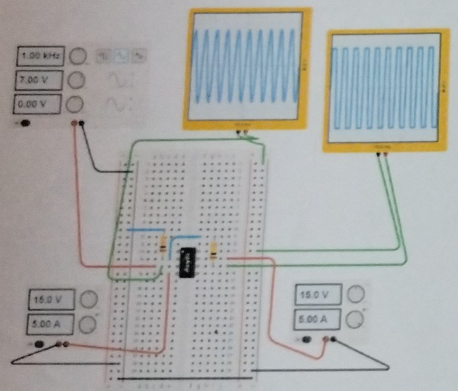
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Inverting Amplifier Using OP-Amp:-



Non-Inverting Amplifier Using OP-AMP:-



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Observation Table :- (Non Inverting Amplifier)

SN	R_f	R	V_i	V_o (Calculated)	V_o (Measured)
1.	10	2.2	0.20	1.109	1.116
2.	13	3.3	0.25	1.386	1.379
3.	22	4.7	0.30	1.704	1.711

Calculation:-

Inverting Amplifier $V_o(Cal) = -(R_f/R) \times V_i = -1.404\,V$

Non Inverting Amplifier (V_o tho) $= [1 + (R_f/R)] \times V_i = 1.704\,V$

% Error between Calculated and Measured V_o = 0.50 % for inverting
= 0.41 % for non inverting.

Conclusion:-

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Based on the Extremly low Calculated error margins (0.50% and 0.41%) the stimulated op-amp circuits behave nearly identically to theoretical voltage output predictions.

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